Gmail **Gowri D <gowri.d.tech@gmail.com>**

Fwd: 8th International Conference on Innovative Trends in Information Technology (ICITIIT 2027) : Submission (1082) has been created.

Prathika <prathieasu@gmail.com> Tue, Oct 13, 2026 at 2:46 PM
To: Abisha Rebekkal <abisharebekkal@gmail.com>, "gowri.d.tech@gmail.com" <gowri.d.tech@gmail.com>

---------- Forwarded message ---------
From: **Microsoft CMT** <noreply@msr-cmt.org>
Date: Tue, Oct 13, 2026 at 2:44 PM
Subject: 8th International Conference on Innovative Trends in Information Technology (ICITIIT 2027) : Submission (1082) has been created.
To: <prathieasu@gmail.com>

Hello,

The following submission has been created.

Track Name: ICITIIT2027

Paper ID: 1082

Paper Title: SmartPCB-XAI: A Dual-Head Deep Learning Framework for Explainable Printed Circuit Board Inspection, Determinitstic Quality Auditing and Digital Twins

Abstract:
Modern-day SMT manufacturing requires AOI technology equipped with precise defect localization on PCBs, simplicity and explanation of decisions. However, today's solutions don't meet these criteria: they need a universal pipeline to localize and classify defects and components, predictions require spatial certainty metrics, quality scores are not related to particular failure modes. We propose a multilayer solution integrating two separated YOLOv11s detectors with their feature activation explanations, IPC-A-610-defined severity score, interactive 2D digital twin and automated generation of reports in PDF and Excel formats. With an independently tested holdout sample of 500 DeepPCB images (3,140 boxes), we achieve 95.11% mAP@50 (63.27% mAP@5095, 92.91% precision, 90.60% recall), with no data leakage, outperforming a recently introduced lightweight model with 92.3%. Our component detector produces only 9.33% mAP@50 on a dataset of 89 images (1,238 boxes) due to class taxonomy misalignment, board bias (64% coming from the same board) and small objects (81.9% under 32px), which is mitigated by per-detection class assignment. End-to-end validation on 20 images was done with no mistakes or unusual zero-detections

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Authors:
- prathieasu@gmail.com (Primary)
- nandhumalli92@gmail.com
- abisharebekkal@gmail.com
- gowri.d.tech@gmail.com

Secondary Subject Areas: Not Entered

Submission Files:
SmartPCB-XAI A Dual-Head Deep Learning Framework for Explainable Printed Circuit Board Inspection, Determinitstic Quality Auditing and Digital Twins.pdf (799 Kb, Tue, 13 Oct 2026 09:14:00 GMT)

Submission Questions Response: Not Entered

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